

TPMS Structures

Stiffness grading optimization

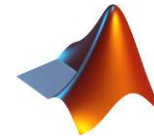


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Outline



► Stiffness grading optimization



Stiffness grading optimization – x-direction

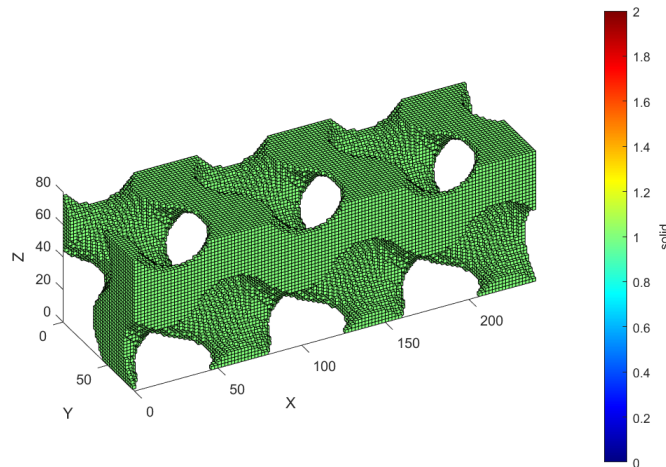
Schoen-Gyroid

$$U_G = \cos(2\pi x) \sin(2\pi y) + \cos(2\pi y) \sin(2\pi z) + \cos(2\pi z) \sin(2\pi x)$$

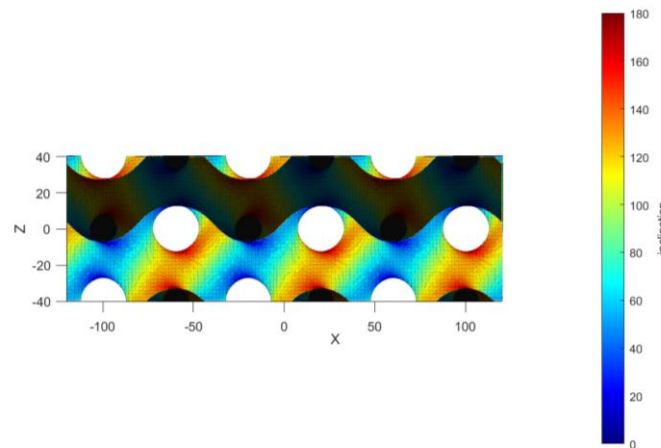
Example

$$U_G = c \quad c = 0.3$$

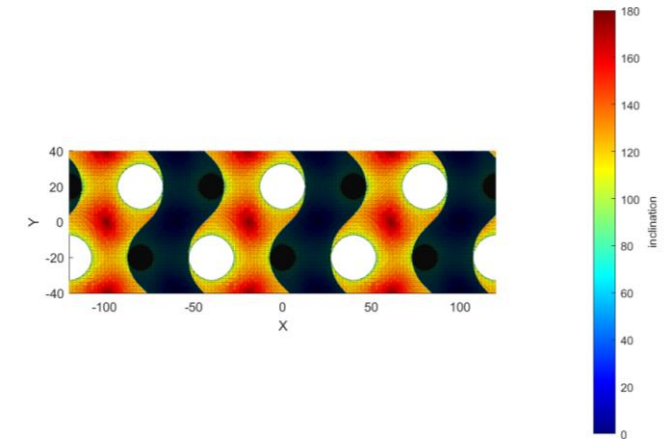
Voxel 3D plot



Surfmesh 3D plot – XZ plane



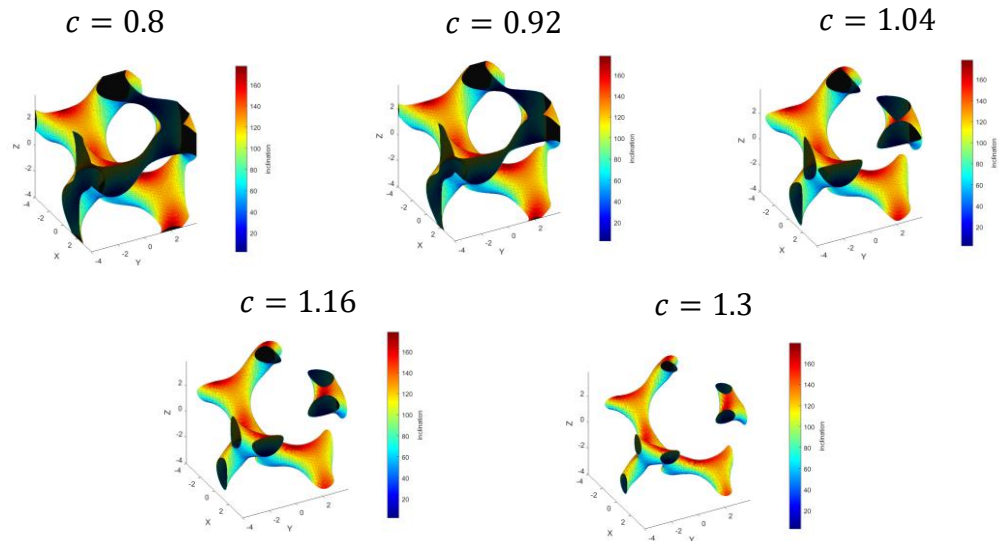
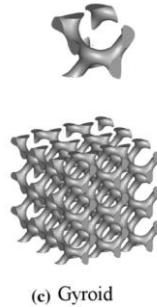
Surfmesh 3D plot – XY plane



Stiffness grading optimization – x-direction

Reference geometry (Abou-Ali2019)

TPMS: Schoen-Gyroid
Unit cell size: 8 mm
Pattern: 5x5x5 cells
Material: polymeric polyamide
Fabrication: laser powder bed fusion-EOS Formiga P110



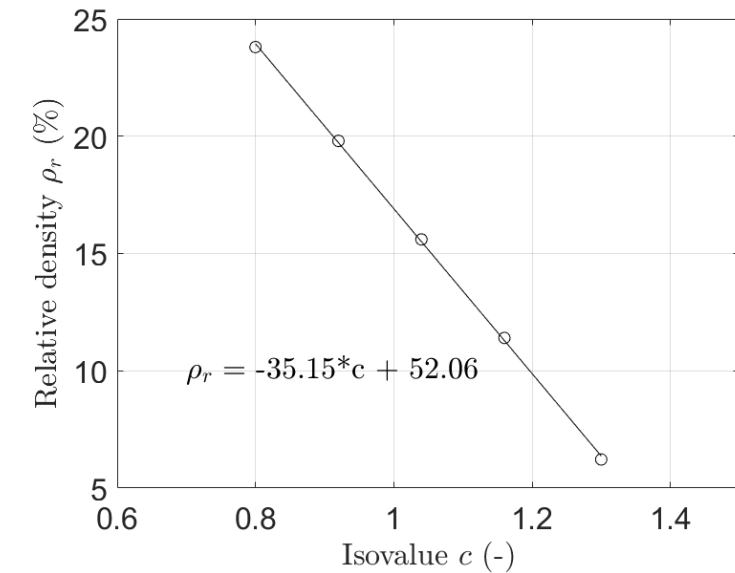
Isovalue-Relative density characterization

Isovalue range: $c = 0.8 - 1.3$

Rel. density range: $\rho_r = 6.1\% - 23.9\%$

Linear fitting

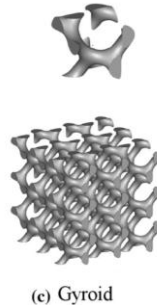
$$\rho_r(c) = a * c + b$$



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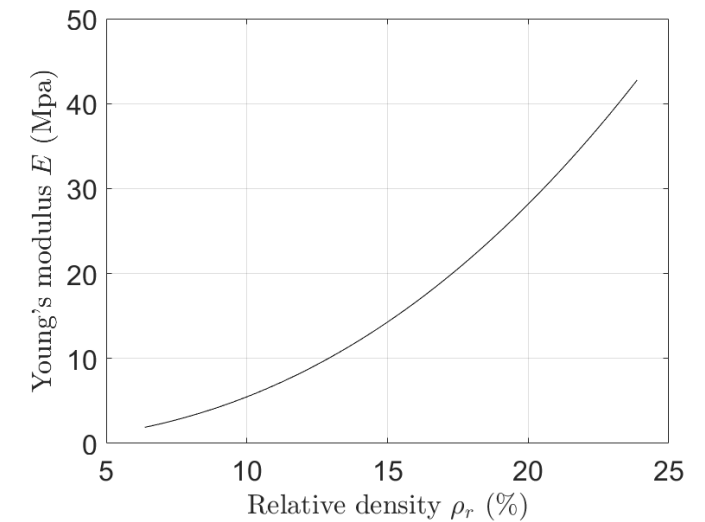
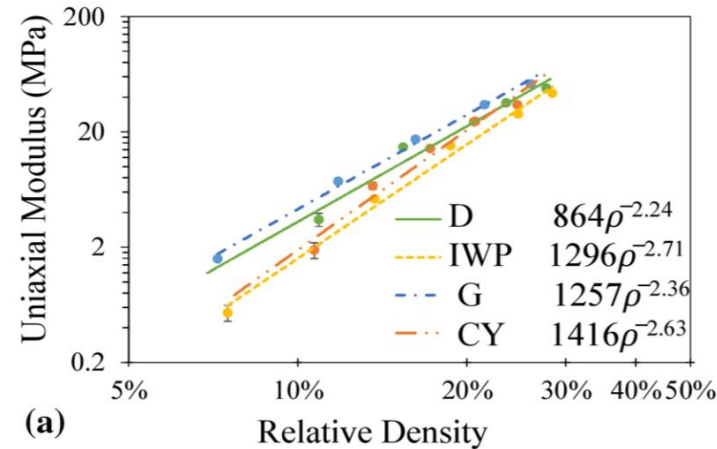


Gibson-Ashby model

$$E(\rho_r) = K\rho_r^n$$

$$K = 1257 \text{ MPa}$$

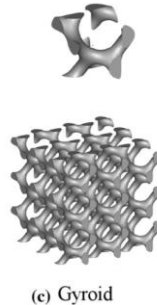
$$n = 2.36$$



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Stiffness optimization

$$E_{opt}(x) = \alpha x + \beta$$

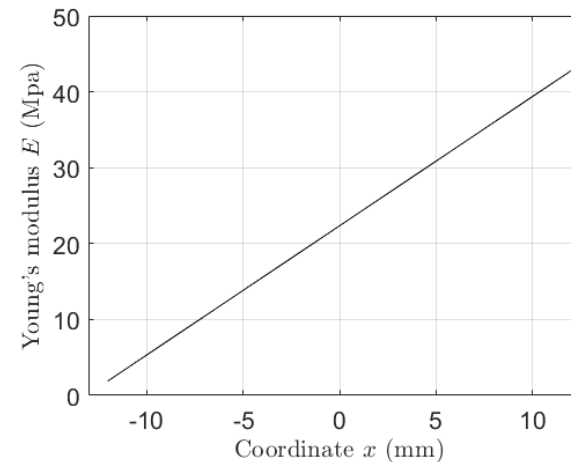
$$\rho_r(c) = a * c + b$$

$$E(\rho_r) = K \rho_r^n$$

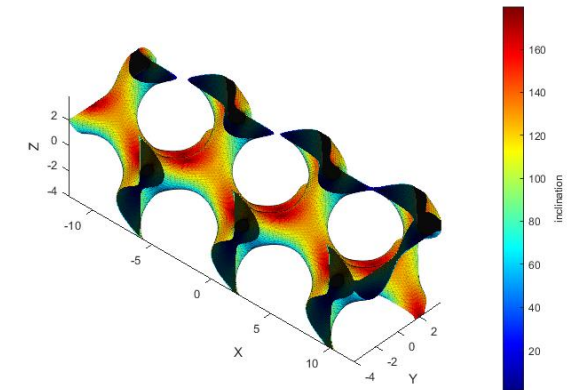
$$\rho_{r,opt}(x) = \left(\frac{\alpha x + \beta}{K} \right)^{1/n}$$

$$c_{opt}(x) = \frac{1}{a} \left(\frac{\alpha x + \beta}{K} \right)^{1/n} - \frac{b}{a}$$

Stiffness target design



Optimized structure



Thanks for your attention



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